University of the Philippines - Baguio

Archie Junio's Reference Notebook

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1. Data Structures

1.1. Segment Tree.

```
class SegmentTree: ------------------------------------------------------------
--- def __init__(self, data): ---------------------------------------------
------- self.n = len(data) ------------------------------------------------
------- self.tree = [0] * (2 * self.n) ------------------------------------
------- for i in range(self.n): -------------------------------------------
----------- self.tree[self.n + i] = data[i] -----------------------------
------- for i in range(self.n - 1, 0, -1): --------------------------------
----------- self.tree[i] = self.tree[i << 1] + self.tree[i << 1 | 1] -
 ------------------------------------------------------------------------
--- def update(self, p, value): -------------------------------------------
------- p += self.n -------------------------------------------------------
------- self.tree[p] = value ----------------------------------------------
------- while p > 1: ------------------------------------------------------
----------- self.tree[p >> 1] = self.tree[p] + self.tree[p ^ 1] ------
----------- p >>= 1 -----------------------------------------------------
 ------------------------------------------------------------------------
--- def query(self, l, r): ------------------------------------------------
------- res = 0 -----------------------------------------------------------
------- l += self.n -------------------------------------------------------
------- r += self.n -------------------------------------------------------
------- while l < r: ------------------------------------------------------
----------- if l & 1: ---------------------------------------------------
--------------- res += self.tree[l] -----------------------------------
--------------- l += 1 ------------------------------------------------
----------- if r & 1: ---------------------------------------------------
--------------- r -= 1 ------------------------------------------------
--------------- res += self.tree[r] -----------------------------------
----------- l >>= 1 -----------------------------------------------------
----------- r >>= 1 -----------------------------------------------------
------- return res --------------------------------------------------------
```

1.2. Union Find.

```
class DSU: --------------------------------------------------------------
--- def __init__(self, n): ------------------------------------------
------- self.parent = list(range(n)) --------------------------------
------- self.size = [1] * n -----------------------------------------
 ------------------------------------------------------------------
--- def find(self, i): ----------------------------------------------
------- if self.parent[i] == i: -------------------------------------
----------- return i ----------------------------------------------
------- self.parent[i] = self.find(self.parent[i]) ------------------
------- return self.parent[i] ---------------------------------------
 ------------------------------------------------------------------
--- def union(self, i, j): ------------------------------------------
------- root_i = self.find(i) ---------------------------------------
------- root_j = self.find(j) ---------------------------------------
------- ------------------------------------------------------------
------- if root_i != root_j: ----------------------------------------
```

```
        if self.size[root_i] < self.size[root_j]:
            root_i, root_j = root_j, root_i
        self.parent[root_j] = root_i
        self.size[root_i] += self.size[root_j]
        return True
    return False

    def get_size(self, i):
        return self.size[self.find(i)]
```

1.3. 2D Sparse Table.

```
import math

class SparseTable2D:
    def __init__(self, A):
        self.A = A
        self.n = len(A)
        self.m = len(A[0])

        self.LOGN = int(math.log2(self.n)) + 1
        self.LOGM = int(math.log2(self.m)) + 1

        self.lg = [0] * (max(self.n, self.m) + 1)
        for i in range(2, len(self.lg)):
            self.lg[i] = self.lg[i // 2] + 1

        self.st = [[[[0] * self.m for _ in range(self.n)]
                   for _ in range(self.LOGM)]
                  for _ in range(self.LOGN)]

        self.build()

    def build(self):
        for i in range(self.n):
            for j in range(self.m):
                self.st[0][0][i][j] = self.A[i][j]

        for bj in range(1, self.LOGM):
            for i in range(self.n):
                for j in range(self.m - (1 << bj) + 1):
                    self.st[0][bj][i][j] = max(
                        self.st[0][bj - 1][i][j],
                        self.st[0][bj - 1][i][j + (1 << (bj - 1))]
                    )

        for bi in range(1, self.LOGN):
            for i in range(self.n - (1 << bi) + 1):
                for j in range(self.m):
                    self.st[bi][0][i][j] = max(
                        self.st[bi - 1][0][i][j],
                        self.st[bi - 1][0][i + (1 << (bi - 1))][j]
                    )

        for bi in range(1, self.LOGN):
            for bj in range(1, self.LOGM):
                for i in range(self.n - (1 << bi) + 1):
                    for j in range(self.m - (1 << bj) + 1):
                        self.st[bi][bj][i][j] = max(
                            self.st[bi - 1][bj - 1][i][j],
                            self.st[bi - 1][bj - 1][i + (1 << (bi - 1))][j],
                            self.st[bi - 1][bj - 1][i][j + (1 << (bj - 1))],
                            self.st[bi - 1][bj - 1][i + (1 << (bi - 1))][j + (1 << (bj - 1))]
                        )

    def query(self, x1, y1, x2, y2):
        kx = self.lg[x2 - x1 + 1]
        ky = self.lg[y2 - y1 + 1]
        return max(
            self.st[kx][ky][x1][y1],
            self.st[kx][ky][x2 - (1 << kx) + 1][y1],
            self.st[kx][ky][x1][y2 - (1 << ky) + 1],
            self.st[kx][ky][x2 - (1 << kx) + 1][y2 - (1 << ky) + 1]
        )
# Input Format: st = SparseTable2D(grid)
#               print(st.query(1, 1, 2, 2))
```

2. Dynamic Programming

2.1. Lichao Tree.

```
class Line:
    __slots__ = ['m', 'c']
    def __init__(self, m=0, c=float('-inf')):
        self.m = m
        self.c = c

    def eval(self, x):
        return self.m * x + self.c

class Node:
    __slots__ = ['line', 'left', 'right']
    def __init__(self):
        self.line = Line()
        self.left = None
        self.right = None

class LiChaoTree:
    def __init__(self, min_x=-10**12, max_x=10**12):
        self.root = Node()
        self.min_x = min_x
        self.max_x = max_x

    def add_line(self, m, c):
        self._insert(self.root, self.min_x, self.max_x, Line(m, c))

    def _insert(self, node, l, r, new_line):
        mid = (l + r) // 2

        left_better = new_line.eval(l) > node.line.eval(l)
        mid_better = new_line.eval(mid) > node.line.eval(mid)

        if mid_better:
            node.line, new_line = new_line, node.line
        if r - l <= 1:
            return

        if left_better != mid_better:
            if not node.left:
                node.left = Node()
            self._insert(node.left, l, mid, new_line)
        else:
            if not node.right:
                node.right = Node()
            self._insert(node.right, mid, r, new_line)

    def query(self, x):
        return self._query(self.root, self.min_x, self.max_x, x)

    def _query(self, node, l, r, x):
        if not node:
            return float('-inf')
        if r - l <= 1:
            return node.line.eval(x)

        mid = (l + r) // 2
        res = node.line.eval(x)

        if x < mid:
            res = max(res, self._query(node.left, l, mid, x))
        else:
            res = max(res, self._query(node.right, mid, r, x))

        return res
```

2.2. Divide And Conquer.

```
INF = float('inf')

def solve_dp(g, k_L, k_R, n_L, n_R, dp, cost_func):
    if n_L > n_R:
        return

    n_M = (n_L + n_R) // 2
    best_k = -1
    dp[g][n_M] = INF

    for k in range(k_L, min(n_M, k_R) + 1):
        current_val = dp[g - 1][k] + cost_func(k + 1, n_M)
        if current_val < dp[g][n_M]:
            dp[g][n_M] = current_val
            best_k = k

    solve_dp(g, k_L, best_k, n_L, n_M - 1, dp, cost_func)
    solve_dp(g, best_k, k_R, n_M + 1, n_R, dp, cost_func)
```

3. Geometry

3.1. Angles And Rotations.

```
def angle(a, b, c): # angle formed by abc in radians: PI < x <= PI
    return abs((arg(a-b) - arg(c-b)) % (2 * math.pi))
def rotate(p, a, d): # rotate point a about pivot p CCW at d radians
    return p + (a - p) * complex(math.cos(d), math.sin(d))
```

3.2. Distances.

```
# Distance: Point to 2D Line (ax + by + c = 0)
def dist_pt_line_eq(p, a, b, c):
    return abs(a*p.real + b*p.imag + c) / math.hypot(a, b)

# Distance: Point to 2D Line through points a and b
def dist_pt_line(p, a, b):
    cp = (p - a).conjugate() * (b - a)
    return abs(cp.imag) / abs(b - a)

# Distance: Point to 3D Plane (ax + by + cz + d = 0)
def dist_pt_plane(p, a, b, c, d):
    # math.hypot supports 3+ arguments in Python 3.8+
    return abs(a*p.x + b*p.y + c*p.z + d) / math.hypot(a,b,c)

# Distance: Shortest distance between 3D lines AB and CD
def dist_line_3d(A, B, C, D):
    u, v, w = B - A, D - C, A - C
    a, b = u.dot(u), u.dot(v)
    c, d, e = v.dot(v), u.dot(w), v.dot(w)
    det = a * c - b * b
    if det < 1e-9: # Parallel lines
        s, t = 0.0, d / b if b > c else e / c
    else: # Skew lines
        s = (b * e - c * d) / det
        t = (a * e - b * d) / det
    p1 = P3(A.x + u.x*s, A.y + u.y*s, A.z + u.z*s)
    p2 = P3(C.x + v.x*t, C.y + v.y*t, C.z + v.z*t)
    vec = p1 - p2
    return math.sqrt(vec.dot(vec))
```

3.3. Intersections.

3.3.1. *Circle Line Intersection.*

```
EPS = 1e-9
def projLine(p, a, b): # Project point p onto line segment ab
    v = b - a
    scalar = ((p - a) * v.conjugate()).real / abs(v)**2
    return a + v * scalar

def rotate(c, p, t): # Rotate point p around center c by t radians
    return c + (p - c) * cmath.rect(1, t)

def CL_inter(c, r, a, b):
    p = projLine(c, a, b)
    d = abs(c - p)
    ans = []

    if d > r + EPS: pass # None
    elif d > r - EPS: ans.append(p) # Tangent
    elif d < EPS: # Diameter
        v = r * (b - a) / abs(b - a)
        ans.append(c + v)
        ans.append(c - v)
    else: # Secant
        t = math.acos(d / r)
        p = c + (p - c) * r / d
        ans.append(rotate(c, p, t))
        ans.append(rotate(c, p, -t))
    return ans
```

3.3.2. *Circle Circle Intersection.*

```
EPS = 1e-9
def rotate(c, p, t):
    # Rotate point p around center c by t radians
    return c + (p - c) * cmath.rect(1, t)

def CC_inter(c1, r1, c2, r2):
    d = abs(c1 - c2)
    ans = []
    if d < EPS: # Concentric circles
        if abs(r1 - r2) < EPS:
            pass # Infinite intersections (same circle)
    elif r1 < EPS: # Circle 1 is a point
        if abs(d - r2) < EPS:
            ans.append(c1)
    else: # General case using Law of Cosines
        s = (r1*r1 + d*d - r2*r2) / (2*r1*d)
        s = max(-1.0, min(1.0, s)) # Prevent float domain errors
        t = math.acos(s)
        # Point on C1's radius pointing towards C2
        mid = c1 + (c2 - c1) * r1 / d
        ans.append(rotate(c1, mid, t))
        if abs(math.sin(t)) >= EPS:
            # FIXED: Rotate around c1, not c2!
            ans.append(rotate(c1, mid, -t))
    return ans
```

3.3.3. *Line Intersect.*

```
def line_inter_fast(a, b, c, d):
    A = cross(d - a, b - a)
    B = cross(c - a, b - a)
    return (B * d - A * c) / (B - A) if B - A else None
```

3.4. Closest Pair.

```
def closest_pair(pts):
    if len(pts) <= 1: return float('inf')
    px = sorted(pts) # Automatically sorts by x, then y
    def solve(l, r): # Base case: Brute force for small partitions
        if r - l <= 3:
            d = float('inf')
            for i in range(l, r):
                for j in range(i + 1, r):
                    d = min(d, math.dist(px[i], px[j]))
            return d
        mid = (l + r) // 2
        mid_x = px[mid][0]
        # Recursively find the minimum in both halves
        d = min(solve(l, mid), solve(mid, r))
        # Gather points in the middle strip
        strip = [p for p in px[l:r] if abs(p[0] - mid_x) < d]
        strip.sort(key=lambda p: p[1]) # Sort strip by y
        # Check at most 7 geometric neighbors in the strip
        for i, p1 in enumerate(strip):
            for p2 in strip[i+1 : i+8]:
                d = min(d, math.dist(p1, p2))
        return d
    return solve(0, len(px))
```

3.5. Convex Hull.

3.5.1. *Convex Hull.*

```
def cross(o, a, b): # 2D cross product of OA and OB vectors
    return (a[0]-o[0])*(b[1]-o[1]) - (a[1]-o[1])*(b[0]-o[0])

def convex_hull(pts): # Sorts by x, then by y automatically
    pts = sorted(set(pts))
    if len(pts) <= 1:
        return pts
    def build(iterable):
        h = []
        for p in iterable:
            while len(h) >= 2 and cross(h[-2], h[-1], p) <= 0:
                h.pop()
            h.append(p)
        return h
    lower = build(pts)
    upper = build(reversed(pts))
    return lower[:-1] + upper[:-1]
```

3.5.2. *Convex Hull 3D.*

```
class P3:
    __slots__ = ['x', 'y', 'z']
    def __init__(self, x=0, y=0, z=0):
        self.x, self.y, self.z = x, y, z
    def __sub__(self, o):
        return P3(self.x-o.x, self.y-o.y, self.z-o.z)
    def cross(self, o):
        return P3(self.y*o.z - self.z*o.y,
                  self.z*o.x - self.x*o.z,
                  self.x*o.y - self.y*o.x)
    def dot(self, o):
        return self.x*o.x + self.y*o.y + self.z*o.z
    def __eq__(self, o):
        return (self.x, self.y, self.z) == (o.x, o.y, o.z)
    def __lt__(self, o):
        return (self.x, self.y, self.z) < (o.x, o.y, o.z)

class Face:
    __slots__ = ['p', 'q']
    def __init__(self, p, q):
        self.p = p # Tuple of 3 point indices
        self.q = q # Normal vector (P3 object)

def hull_3D(pts):
    n = len(pts)
    faces = []
    dead = [[True] * n for _ in range(n)]
    def add_face(a, b, c):
        q = (pts[b] - pts[a]).cross(pts[c] - pts[a])
        faces.append(Face((a, b, c), q))
        dead[a][b] = dead[b][c] = dead[c][a] = False
    # Initialize with a degenerate tetrahedron
    add_face(0, 1, 2)
    add_face(0, 2, 1)
    for i in range(3, n):
        f_inv = []
        for f in faces:
            # If face is visible from the point
            if (pts[i] - pts[f.p[0]]).dot(f.q) > 0:
                for j in range(3):
                    dead[f.p[j]][f.p[(j+1)%3]] = True
            else:
                f_inv.append(f)
        faces = []
        for f in f_inv:
            for j in range(3):
                a, b = f.p[j], f.p[(j+1)%3]
                # If reverse edge is dead but forward is alive
                if dead[b][a]:
                    add_face(b, a, i)
        faces.extend(f_inv)
    return faces
```

3.6. Areas.

3.6.1. *Triangle Area.*

```
def area(a, b, c):
    s = (a + b + c) / 2
    return (s*(s-a)*(s-b)*(s-c)) ** 0.5
```

3.6.2. *Polygon Area.*

```
def area(p, n):
    a = 0
    for i in range(n):
        j = (i - 1) % n
        a += cross(p[i], p[j])
    return abs(a) / 2
```

3.7. Dot Cross.

```
def dot(a, b):
    return a.x * b.x + a.y * b.y # + a.z * b.z
def cross(a, b):
    return a.x * b.y - a.y * b.x
def cross(a, b, c):
    return cross(a, b) + cross(b, c) + cross(c, a)
def cross3D(a, b):
    return point3(a.x*b.y - a.y*b.x, a.y*b.z - a.z*b.y, a.z*b.x -
↪ a.x*b.z)
```

3.8. Line Sweep Closest Pair.

```
import math

def dist_sq(p1, p2):
    return (p1[0] - p2[0])**2 + (p1[1] - p2[1])**2

def closest_pair(points):
    points.sort()

    def solve(pts):
        n = len(pts)

        if n <= 3:
            return min(dist_sq(pts[i], pts[j])
                       for i in range(n)
                       for j in range(i + 1, n)) if n > 1 else
↪ float('inf')

        mid = n // 2
        mid_x = pts[mid][0]

        left = pts[:mid]
```

```
------- right = pts[mid:]
------- d = min(solve(left), solve(right))
------- strip = [p for p in pts if (p[0] - mid_x) ** 2 < d]
------- strip.sort(key=lambda p: p[1])
------- for i in range(len(strip)):
----------- for j in range(i + 1, min(i + 7, len(strip))):
--------------- d = min(d, dist_sq(strip[i], strip[j]))
------- return d
--- return solve(points)
```

3.9. Point Projection.

```
EPS = 1e-9

def dot(a, b):
--- return sum(x * y for x, y in zip(a, b))


def scale(v, k):
--- return [k * x for x in v]


def add(a, b):
--- return [x + y for x, y in zip(a, b)]


def sub(a, b):
--- return [x - y for x, y in zip(a, b)]


def norm(v):
--- return dot(v, v)

def proj(p, v):
--- v_norm = norm(v)
--- if v_norm < EPS:
------- return [0] * len(p)
--- return scale(v, dot(p, v) / v_norm)

def proj_line(p, a, b):
--- ab = sub(b, a)
--- ab_norm = norm(ab)
--- if ab_norm < EPS:
------- return a[:]

--- t = dot(sub(p, a), ab) / ab_norm
--- return add(a, scale(ab, t))

def proj_seg(p, a, b):
--- ab = sub(b, a)
--- ab_norm = norm(ab)
--- if ab_norm < EPS:
------- return a[:]

--- t = dot(sub(p, a), ab) / ab_norm
--- t = max(0.0, min(1.0, t))

--- return add(a, scale(ab, t))

def proj_plane(p, a, b, c, d):
--- n = [a, b, c]
--- n_norm = norm(n)
--- if n_norm < EPS:
------- return p[:]

--- dist = (dot(p, n) + d) / n_norm
--- return sub(p, scale(n, dist))
```

3.10. Great Circle Distance.

```
import math

PI = math.pi

def clamp(x, lo, hi):
--- return max(lo, min(hi, x))

# ========================================================
# version 1: lat/long (spherical law of cosines)
# ========================================================
def great_circle_dist_latlong(lat1, lon1, lat2, lon2, R=6371.0):
--- lat1 *= PI / 180
--- lon1 *= PI / 180
--- lat2 *= PI / 180
--- lon2 *= PI / 180

--- val = (
------- math.sin(lat1) * math.sin(lat2)
------- + math.cos(lat1) * math.cos(lat2) * math.cos(abs(lon1 - lon2))
--- )

--- val = clamp(val, -1.0, 1.0)  # IMPORTANT FIX

--- return R * math.acos(val)

def dot(a, b):
--- return sum(x * y for x, y in zip(a, b))

def cross(a, b):
--- return [
------- a[1] * b[2] - a[2] * b[1],
------- a[2] * b[0] - a[0] * b[2],
------- a[0] * b[1] - a[1] * b[0],
--- ]

def norm(v):
--- return math.sqrt(dot(v, v))

# ========================================================
# version 2: using 3D vectors (angle between vectors)
# ========================================================
def great_circle_dist_vec(a, b, R=1.0):
--- # assumes a and b are unit vectors on sphere
--- cross_prod = cross(a, b)
--- return R * math.atan2(norm(cross_prod), dot(a, b))

def to_vec(lat, lon):
--- lat *= PI / 180
--- lon *= PI / 180

--- return [
------- math.cos(lat) * math.cos(lon),
------- math.cos(lat) * math.sin(lon),
------- math.sin(lat),
--- ]
```

3.11. Point In Polygon.

```
import math

EPS = 1e-9

def dist(a, b):
--- return math.hypot(a[0] - b[0], a[1] - b[1])

def in_polygon(q, p):
--- n = len(p)
--- inside = False

--- j = n - 1
--- for i in range(n):
------- xi, yi = p[i]
------- xj, yj = p[j]
------- xq, yq = q

------- intersect = ((yi > yq) != (yj > yq)) and (
----------- xq < (xj - xi) * (yq - yi) / (yj - yi + EPS) + xi
------- )

------- if intersect:
----------- inside = not inside

------- j = i

--- return inside

def on_polygon(q, p):
--- n = len(p)

--- for i in range(n):
------- j = (i + 1) % n

------- a = p[i]
------- b = p[j]

------- if abs(dist(a, q) + dist(q, b) - dist(a, b)) < EPS:
----------- return True

--- return False
```

3.12. Cut Polygon By Line.

```
EPS = 1e-9

def cross(a, b, c):
--- return (b[0] - a[0]) * (c[1] - a[1]) - (b[1] - a[1]) * (c[0] -
↪  a[0])

def line_inter(p1, p2, a, b):
--- x1, y1 = p1
--- x2, y2 = p2
--- x3, y3 = a
--- x4, y4 = b

--- den = (x1 - x2) * (y3 - y4) - (y1 - y2) * (x3 - x4)
--- if abs(den) < EPS:
------- return p1

--- px = (
------- (x1 * y2 - y1 * x2) * (x3 - x4)
------- - (x1 - x2) * (x3 * y4 - y3 * x4)
--- ) / den

--- py = (
------- (x1 * y2 - y1 * x2) * (y3 - y4)
------- - (y1 - y2) * (x3 * y4 - y3 * x4)
--- ) / den

--- return (px, py)

def cut(poly, a, b):
--- n = len(poly)
--- res = []

--- for i in range(n):
------- j = (i - 1 + n) % n

------- c1 = cross(a, b, poly[j])
------- c2 = cross(a, b, poly[i])

------- if c1 >= -EPS:
```

```
----------- res.append(poly[j]) ----------------------------------------
-------------------------------------------------------------------------
------- if c1 * c2 < -EPS: ----------------------------------------------
----------- inter = line_inter(poly[j], poly[i], a, b) ----------------
----------- res.append(inter) -----------------------------------------
-------------------------------------------------------------------------
--- return res ------------------------------------------------------------
```

4. Graphs

4.1. Max Flow.

4.1.1. *Dinic.*

```
INF = float('inf') ----------------------------------------------------
class Dinic: ----------------------------------------------------------
--- def __init__(self, n, s, t): ------------------------------------
------- self.n, self.s, self.t = n, s, t ------------------------------
------- self.edges = []  ----------------------------------------------
------- self.adj = [[] for _ in range(n)] -----------------------------
-------------------------------------------------------------------------
--- def add_edge(self, u, v, c, bi=False): --------------------------
------- self.adj[u].append(len(self.edges)) ---------------------------
------- self.edges.append([v, c, 0]) ----------------------------------
------- self.adj[v].append(len(self.edges)) ---------------------------
------- self.edges.append([u, c if bi else 0, 0]) ---------------------
-------------------------------------------------------------------------
--- def bfs(self): --------------------------------------------------
------- self.dist = [-1] * self.n -------------------------------------
------- self.dist[self.s] = 0 -----------------------------------------
------- q = deque([self.s]) -------------------------------------------
------- while q: ------------------------------------------------------
----------- u = q.popleft() -----------------------------------------
----------- for i in self.adj[u]: -----------------------------------
--------------- v, c, f = self.edges[i] -------------------------------
--------------- if self.dist[v] < 0 and c > f: ------------------------
------------------- self.dist[v] = self.dist[u] + 1 -------------------
------------------- q.append(v) ---------------------------------------
------- return self.dist[self.t] != -1 --------------------------------
---  ------------------------------------------------------------------
--- def dfs(self, u, pushed): ---------------------------------------
------- if pushed == 0 or u == self.t: return pushed ------------------
------- while self.ptr[u] < len(self.adj[u]): -------------------------
----------- i = self.adj[u][self.ptr[u]] ----------------------------
----------- v, c, f = self.edges[i] ---------------------------------
----------- if self.dist[u]+1 == self.dist[v] and c > f: ------------
--------------- tr = self.dfs(v, min(pushed, c - f)) ------------------
--------------- if tr > 0: --------------------------------------------
------------------- self.edges[i][2] += tr ----------------------------
------------------- self.edges[i^1][2] -= tr --------------------------
------------------- return tr -----------------------------------------
----------- self.ptr[u] += 1 ----------------------------------------
------- return 0 ------------------------------------------------------
-------------------------------------------------------------------------
--- def calc_max_flow(self): ----------------------------------------
------- flow = 0 ------------------------------------------------------
------- while self.bfs(): ---------------------------------------------
----------- self.ptr = [0] * self.n ---------------------------------
----------- while True: ---------------------------------------------
--------------- pushed = self.dfs(self.s, INF) ------------------------
--------------- if not pushed: ----------------------------------------
------------------- break ---------------------------------------------
--------------- flow += pushed ----------------------------------------
------- return flow ---------------------------------------------------
```

4.1.2. *Edmonds Karp.*

```
INF = float('inf') ----------------------------------------------------
class EdmondsKarp: ----------------------------------------------------
--- def __init__(self, n, s, t): ------------------------------------
------- self.n, self.s, self.t = n, s, t ------------------------------
------- self.adj = [[] for _ in range(n)] -----------------------------
------- self.res = [[0] * n for _ in range(n)] ------------------------
-------------------------------------------------------------------------
--- def add_edge(self, u, v, w): ------------------------------------
------- self.adj[u].append(v) -----------------------------------------
------- self.adj[v].append(u) -----------------------------------------
------- self.res[u][v] += w -------------------------------------------
-------------------------------------------------------------------------
--- def bfs(self): --------------------------------------------------
------- self.par = [-1] * self.n --------------------------------------
------- self.par[self.s] = self.s -------------------------------------
------- q = deque([self.s]) -------------------------------------------
------- while q: ------------------------------------------------------
----------- u = q.popleft() -----------------------------------------
----------- for v in self.adj[u]: -----------------------------------
--------------- if self.res[u][v] > 0 and self.par[v] == -1: ---------
------------------- self.par[v] = u -----------------------------------
------------------- if v == self.t: -----------------------------------
----------------------- return True -------------------------------------
------------------- q.append(v) ---------------------------------------
------- return False --------------------------------------------------
-------------------------------------------------------------------------
--- def calc_max_flow(self): ----------------------------------------
------- ans = 0 -------------------------------------------------------
------- while self.bfs(): ---------------------------------------------
----------- flow = INF ----------------------------------------------
----------- u = self.t ----------------------------------------------
----------- while u != self.s: --------------------------------------
--------------- p = self.par[u] ---------------------------------------
--------------- flow = min(flow, self.res[p][u]) ----------------------
--------------- u = p -------------------------------------------------
----------- u = self.t ----------------------------------------------
----------- while u != self.s: --------------------------------------
--------------- p = self.par[u] ---------------------------------------
--------------- self.res[p][u] -= flow --------------------------------
--------------- self.res[u][p] += flow --------------------------------
--------------- u = p -------------------------------------------------
----------- ans += flow ---------------------------------------------
------- return ans ----------------------------------------------------
```

4.1.3. *Bipartite Matching.*

```
from collections import deque -----------------------------------------
-------------------------------------------------------------------------
INF = float('inf') ----------------------------------------------------
class BipartiteGraph: -------------------------------------------------
--- def __init__(self, n, m): ---------------------------------------
------- self.n = n ----------------------------------------------------
------- self.m = m ----------------------------------------------------
------- self.adj = [[] for _ in range(n)] -----------------------------
------- self.L = [-1] * n ---------------------------------------------
------- self.R = [-1] * m ---------------------------------------------
------- self.dist = [INF] * (n + 1) -----------------------------------
-------------------------------------------------------------------------
--- def add_edge(self, u, v): ---------------------------------------
------- self.adj[u].append(v) -----------------------------------------
-------------------------------------------------------------------------
--- def bfs(self): --------------------------------------------------
------- queue = deque() -----------------------------------------------
------- for v in range(self.n): ---------------------------------------
----------- if self.L[v] == -1: -------------------------------------
--------------- self.dist[v] = 0 --------------------------------------
--------------- queue.append(v) ---------------------------------------
----------- else: ---------------------------------------------------
--------------- self.dist[v] = INF ------------------------------------
-------------------------------------------------------------------------
------- self.dist[self.n] = INF ---------------------------------------
-------------------------------------------------------------------------
------- while queue: --------------------------------------------------
----------- v = queue.popleft() -------------------------------------
----------- if self.dist[v] < self.dist[self.n]: --------------------
--------------- for u in self.adj[v]: --------------------------------
------------------- next_v = self.R[u] --------------------------------
------------------- dist_next = self.dist[next_v] if next_v != -1 else
↪  self.dist[self.n]
-------------------------------------------------------------------------
------------------- if (next_v == -1 and self.dist[self.n] == INF) or \
----------------------------- (next_v != -1 and self.dist[next_v] ==
↪  INF):
-------------------------------------------------------------------------
----------------------- target = next_v if next_v != -1 else self.n --
----------------------- self.dist[target] = self.dist[v] + 1 ---------
----------------------- if target != self.n: ------------------------
----------------------------- queue.append(target) ------------------
-------------------------------------------------------------------------
------- return self.dist[self.n] != INF -------------------------------
-------------------------------------------------------------------------
--- def dfs(self, v): -----------------------------------------------
------- if v != -1: ---------------------------------------------------
----------- for u in self.adj[v]: -----------------------------------
--------------- next_v = self.R[u] ------------------------------------
--------------- dist_next = self.dist[next_v] if next_v != -1 else
↪  self.dist[self.n]
-------------------------------------------------------------------------
--------------- if dist_next == self.dist[v] + 1: ---------------------
------------------- if self.dfs(next_v): ------------------------------
----------------------- self.R[u] = v -----------------------------------
----------------------- self.L[v] = u -----------------------------------
----------------------- return True -------------------------------------
----------- self.dist[v] = INF --------------------------------------
----------- return False --------------------------------------------
------- return True ---------------------------------------------------
-------------------------------------------------------------------------
--- def maximum_matching(self): -------------------------------------
------- matching = 0 --------------------------------------------------
------- for u in range(self.n): ---------------------------------------
----------- for v in self.adj[u]: -----------------------------------
--------------- if self.R[v] == -1: ----------------------------------
------------------- self.L[u] = v -------------------------------------
------------------- self.R[v] = u -------------------------------------
------------------- matching += 1 -------------------------------------
------------------- break ---------------------------------------------
-------------------------------------------------------------------------
------- while self.bfs(): ---------------------------------------------
----------- for i in range(self.n): ---------------------------------
--------------- if self.L[i] == -1 and self.dfs(i): ------------------
------------------- matching += 1 -------------------------------------
------- return matching -----------------------------------------------
# Input Format: g = BipartiteGraph(2, 2) ----------------------------
#               g.add_edge(0, 0) ------------------------------------
#               g.add_edge(0, 1) ------------------------------------
#               g.add_edge(1, 0) ------------------------------------
#               print(f"Max Matching: {g.maximum_matching()}") -------
```

4.2. Shortest Path.

4.2.1. *Dijkstra.*

```
import heapq ----------------------------------------------------------
INF = float('inf') ----------------------------------------------------
def dijkstra(n, adj, s): ----------------------------------------------
--- dist = [INF] * n ------------------------------------------------
--- dist[s] = 0 -----------------------------------------------------
--- pq = [(0, s)] ---------------------------------------------------
---  ----------------------------------------------------------------
--- while pq: -------------------------------------------------------
------- d, u = heapq.heappop(pq) --------------------------------------
------- if dist[u] < d: -----------------------------------------------
----------- continue ------------------------------------------------
------- for v, w in adj[u]: -------------------------------------------
----------- if dist[v] > d + w: -------------------------------------
--------------- dist[v] = d + w ---------------------------------------
--------------- heapq.heappush(pq, (dist[v], v)) ----------------------
--- return dist -----------------------------------------------------
```

4.2.2. *Bellman Ford.*

```
INF = float('inf')
def bellman_ford(n, adj, s):
    dist = [INF] * n
    dist[s] = 0
    for _ in range(n - 1):
        for u in range(n):
            if dist[u] == INF: continue
            for v, w in adj[u]:
                if dist[u] + w < dist[v]:
                    dist[v] = dist[u] + w
    for u in range(n):
        if dist[u] == INF: continue
        for v, w in adj[u]:
            if dist[u] + w < dist[v]:
                return dist, True
    return dist, False
```

4.3. Minimum Spanning Tree.

4.3.1. *Kruskal Min Spanning Tree.*

```
import heapq

class UnionFind:
    def __init__(self, n):
        self.parent = list(range(n))
        self.rank = [0] * n

    def find(self, x):
        while self.parent[x] != x:
            self.parent[x] = self.parent[self.parent[x]]
            x = self.parent[x]
        return x

    def unite(self, a, b):
        a = self.find(a)
        b = self.find(b)

        if a == b:
            return False

        if self.rank[a] < self.rank[b]:
            self.parent[a] = b
        else:
            self.parent[b] = a
            if self.rank[a] == self.rank[b]:
                self.rank[a] += 1

        return True


def kruskal(n, edges):
    edges = edges[:]
    heapq.heapify(edges)

    uf = UnionFind(n)
    total_weight = 0
    mst = []

    while edges:
        w, u, v = heapq.heappop(edges)
        if uf.unite(u, v):
            mst.append((u, v, w))
            total_weight += w

            if len(mst) == n - 1:
                break

    return mst, total_weight
```

4.3.2. *Prim Min Spanning Tree.*

```
import heapq

def prim(n, adj, start=0):
    visited = [False] * n
    pq = [(0, start, -1)]

    total_weight = 0
    mst = []

    while pq:
        w, u, parent = heapq.heappop(pq)

        if visited[u]:
            continue

        visited[u] = True

        if parent != -1:
            mst.append((parent, u, w))
            total_weight += w

        for v, weight in adj[u]:
            if not visited[v]:
                heapq.heappush(pq, (weight, v, u))

    return mst, total_weight
```

4.4. Least Common Ancestor.

4.4.1. *Binary Lifting Lca.*

```
import math

class BinaryLifting:
    def __init__(self, adj):
        self.n = len(adj)
        self.LOG = math.ceil(math.log2(self.n)) + 1
        self.adj = adj

        self.dep = [0] * self.n
        self.par = [[-1] * self.LOG for _ in range(self.n)]

        self.build(0)

    def dfs(self, u, p):
        self.par[u][0] = p
        for v in self.adj[u]:
            if v != p:
                self.dep[v] = self.dep[u] + 1
                self.dfs(v, u)

    def build(self, root):
        self.dfs(root, root)

        for k in range(1, self.LOG):
            for u in range(self.n):
                self.par[u][k] = self.par[self.par[u][k-1]][k-1]

    def lift(self, u, k):
        for i in range(self.LOG):
            if k & (1 << i):
                u = self.par[u][i]
        return u

    def lca(self, u, v):
        if self.dep[u] < self.dep[v]:
            u, v = v, u

        u = self.lift(u, self.dep[u] - self.dep[v])

        if u == v:
            return u

        for k in reversed(range(self.LOG)):
            if self.par[u][k] != self.par[v][k]:
                u = self.par[u][k]
                v = self.par[v][k]

        return self.par[u][0]
```

4.4.2. *Euler Lca.*

```
class EulerLCA:
    def __init__(self, adj):
        self.n = len(adj)
        self.adj = adj

        self.euler = []
        self.first = [-1] * self.n
        self.depth = [0] * self.n

        self.build(0)

    def dfs(self, u, p, d):
        self.first[u] = len(self.euler)
        self.depth[u] = d
        self.euler.append(u)

        for v in self.adj[u]:
            if v != p:
                self.dfs(v, u, d + 1)
                self.euler.append(u)

    def build(self, root):
        self.dfs(root, root, 0)

        m = len(self.euler)
        LOG = m.bit_length()

        self.st = [[0] * m for _ in range(LOG)]
        self.log = [0] * (m + 1)

        for i in range(2, m + 1):
            self.log[i] = self.log[i // 2] + 1

        for i in range(m):
            self.st[0][i] = self.euler[i]

        for k in range(1, LOG):
            for i in range(m - (1 << k) + 1):
                a = self.st[k-1][i]
                b = self.st[k-1][i + (1 << (k-1))]
                self.st[k][i] = a if self.depth[a] < self.depth[b] else
↪ b

    def lca(self, u, v):
        l, r = self.first[u], self.first[v]
        if l > r:
            l, r = r, l

        k = self.log[r - l + 1]
        a = self.st[k][l]
        b = self.st[k][r - (1 << k) + 1]

        return a if self.depth[a] < self.depth[b] else b
```

4.4.3. *Tarjan Lca.*

```
class DSU:
    def __init__(self, n):
        self.p = list(range(n))

    def find(self, x):
        if self.p[x] != x:
```

```
        self.p[x] = self.find(self.p[x])
        return self.p[x]

    def unite(self, a, b):
        self.p[self.find(b)] = self.find(a)

class Tarjan:
    def __init__(self, adj, queries):
        self.adj = adj
        self.queries = queries
        self.n = len(adj)

        self.dsu = DSU(self.n)
        self.vis = [False] * self.n
        self.anc = [0] * self.n
        self.res = {}

    def dfs(self, u, p):
        self.anc[u] = u

        for v in self.adj[u]:
            if v == p:
                continue
            self.dfs(v, u)
            self.dsu.unite(u, v)
            self.anc[self.dsu.find(u)] = u

        self.vis[u] = True

        for v, idx in self.queries[u]:
            if self.vis[v]:
                self.res[idx] = self.anc[self.dsu.find(v)]
```

4.5. Minimum Mean Weight Cycle.

```
def min_mean_cycle(adj):
    n = len(adj)
    INF = float('inf')

    arr = [[INF] * n for _ in range(n + 1)]

    for i in range(n):
        arr[0][i] = 0

    for k in range(1, n + 1):
        for u in range(n):
            if arr[k - 1][u] == INF:
                continue
            for v, w in adj[u]:
                arr[k][v] = min(arr[k][v], arr[k - 1][u] + w)

    result = INF

    for v in range(n):
        if arr[n][v] == INF:
            continue

        max_avg = -INF
        for k in range(n):
            if arr[k][v] == INF:
                continue
            avg = (arr[n][v] - arr[k][v]) / (n - k)
            max_avg = max(max_avg, avg)

        result = min(result, max_avg)

    return result
```

4.6. Strongly Connected Components.

```
class KosarajuGraph:
    def __init__(self, n):
        self.n = n
        self.graph = [[] for _ in range(n)]
        self.rev_graph = [[] for _ in range(n)]
        self.vis = [0] * n
        self.sccs = []

    def add_edge(self, u, v):
        self.graph[u].append(v)
        self.rev_graph[v].append(u)

    def dfs(self, u, adj, container):
        self.vis[u] = 1
        for v in adj[u]:
            if not self.vis[v]:
                self.dfs(v, adj, container)
        container.append(u)

    def kosaraju(self):
        topo = []

        # First pass
        self.vis = [0] * self.n
        for i in range(self.n):
            if not self.vis[i]:
                self.dfs(i, self.graph, topo)

        # Second pass
        self.vis = [0] * self.n
        self.sccs = []

        for u in reversed(topo):
            if not self.vis[u]:
                comp = []
                self.dfs(u, self.rev_graph, comp)
                self.sccs.append(comp)

        return self.sccs
```

4.7. Heavy Light Decomposition.

```
class SegmentTree:
    def __init__(self, n):
        self.n = n
        self.tree = [0] * (4 * n)

    # Example: point update
    def update(self, idx, val, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if l == r:
            self.tree[node] = vala
            return
        mid = (l + r) // 2
        if idx <= mid:
            self.update(idx, val, node * 2, l, mid)
        else:
            self.update(idx, val, node * 2 + 1, mid + 1, r)
        self.tree[node] = self.tree[node * 2] + self.tree[node * 2 + 1]

    # Example: range query
    def query(self, ql, qr, node=1, l=0, r=None):
        if r is None:
            r = self.n - 1
        if qr < l or r < ql:
            return 0
        if ql <= l and r <= qr:
            return self.tree[node]
        mid = (l + r) // 2
        return self.query(ql, qr, node * 2, l, mid) + \
               self.query(ql, qr, node * 2 + 1, mid + 1, r)


class HeavyLightTree:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.par = [-1] * n
        self.heavy = [-1] * n
        self.dep = [0] * n
        self.path_root = [0] * n
        self.pos = [0] * n

        self.segment_tree = SegmentTree(n)
        self.current_pos = 0

    def add_edge(self, u, v):
        self.adj[u].append(v)
        self.adj[v].append(u)

    def dfs(self, u):
        size = 1
        max_subtree_size = 0

        for v in self.adj[u]:
            if v != self.par[u]:
                self.par[v] = u
                self.dep[v] = self.dep[u] + 1
                subtree_size = self.dfs(v)

                if subtree_size > max_subtree_size:
                    max_subtree_size = subtree_size
                    self.heavy[u] = v

                size += subtree_size

        return size

    def decompose(self, u, root):
        self.path_root[u] = root
        self.pos[u] = self.current_pos
        self.current_pos += 1

        if self.heavy[u] != -1:
            self.decompose(self.heavy[u], root)

        for v in self.adj[u]:
            if v != self.par[u] and v != self.heavy[u]:
                self.decompose(v, v)

    def build(self, root=0):
        self.par[root] = root
        self.dep[root] = 0
        self.dfs(root)
        self.decompose(root, root)

    # Query path u -> v
    def query(self, u, v):
        res = 0
        while self.path_root[u] != self.path_root[v]:
            if self.dep[self.path_root[u]] <
↪   self.dep[self.path_root[v]]:
                u, v = v, u
            start = self.path_root[u]
            res += self.segment_tree.query(self.pos[start],
↪   self.pos[u])
            u = self.par[start]

        # Same path
        if self.dep[u] > self.dep[v]:
            u, v = v, u
        res += self.segment_tree.query(self.pos[u], self.pos[v])
        return res
```

```
--- # Update node value
--- def update(self, u, value):
------- self.segment_tree.update(self.pos[u], value)
```

4.8. Euler Path/Cycle.

4.8.1. *Directed Euler.*

```
from collections import defaultdict

class DirectedEuler:
--- def __init__(self, n):
------- self.n = n
------- self.adj = defaultdict(list)
------- self.indeg = [0]*n
------- self.outdeg = [0]*n
------- self.edge_count = 0

--- def add_edge(self, u, v):
------- self.adj[u].append(v)
------- self.outdeg[u] += 1
------- self.indeg[v] += 1
------- self.edge_count += 1

--- def find_start_node(self):
------- start = -1
------- end = -1
------- for i in range(self.n):
----------- if self.outdeg[i] - self.indeg[i] == 1:
--------------- if start != -1:
------------------- return -1
--------------- start = i
----------- elif self.indeg[i] - self.outdeg[i] == 1:
--------------- if end != -1:
------------------- return -1
--------------- end = i
----------- elif self.indeg[i] != self.outdeg[i]:
--------------- return -1

------- if start == -1:
----------- for i in range(self.n):
--------------- if self.outdeg[i] > 0:
------------------- return i
------- return start

--- def get_euler_path(self):
------- start = self.find_start_node()
------- if start == -1:
----------- return None

------- stack = [start]
------- path = []

------- local_adj = {u: self.adj[u][:] for u in self.adj}

------- while stack:
----------- u = stack[-1]
----------- if u in local_adj and local_adj[u]:
--------------- v = local_adj[u].pop()
--------------- stack.append(v)
----------- else:
--------------- path.append(stack.pop())

------- if len(path) != self.edge_count + 1:
----------- return None

------- return path[::-1]

--- class UndirectedEuler:
------- def __init__(self, n):
----------- self.n = n
----------- self.adj = defaultdict(list)
----------- self.edge_count = 0

------- def add_edge(self, u, v):
----------- self.adj[u].append(v)
----------- self.adj[v].append(u)
----------- self.edge_count += 1

------- def is_valid(self):
----------- odd = 0
----------- for i in range(self.n):
--------------- if len(self.adj[i]) % 2 != 0:
------------------- odd += 1
----------- return odd == 0 or odd == 2

------- def get_euler_path(self):
----------- if not self.is_valid():
--------------- return None

----------- start = 0
----------- for i in range(self.n):
--------------- if len(self.adj[i]) % 2 == 1:
------------------- start = i
------------------- break
--------------- if len(self.adj[i]) > 0:
------------------- start = i

----------- stack = [start]
----------- path = []

----------- local_adj = {u: self.adj[u][:] for u in self.adj}

----------- while stack:
--------------- u = stack[-1]
--------------- if u in local_adj and local_adj[u]:
------------------- v = local_adj[u].pop()
------------------- local_adj[v].remove(u)
------------------- stack.append(v)
--------------- else:
------------------- path.append(stack.pop())

----------- if len(path) != self.edge_count + 1:
--------------- return None

----------- return path[::-1]
```

4.8.2. *Undirected Euler.*

```
from collections import defaultdict

class UndirectedEuler:
--- def __init__(self, n):
------- self.n = n
------- self.adj = defaultdict(list)
------- self.edge_count = 0

--- def add_edge(self, u, v):
------- self.adj[u].append(v)
------- self.adj[v].append(u)
------- self.edge_count += 1

--- def is_valid(self):
------- odd = 0
------- for i in range(self.n):
----------- if len(self.adj[i]) % 2 != 0:
--------------- odd += 1
------- return odd == 0 or odd == 2

--- def get_euler_path(self):
------- if not self.is_valid():
----------- return None

------- start = 0
------- for i in range(self.n):
----------- if len(self.adj[i]) % 2 == 1:
--------------- start = i
--------------- break
----------- if len(self.adj[i]) > 0:
--------------- start = i

------- stack = [start]
------- path = []

------- local_adj = {u: self.adj[u][:] for u in self.adj}

------- while stack:
----------- u = stack[-1]
----------- if u in local_adj and local_adj[u]:
--------------- v = local_adj[u].pop()
--------------- local_adj[v].remove(u)
--------------- stack.append(v)
----------- else:
--------------- path.append(stack.pop())

------- if len(path) != self.edge_count + 1:
----------- return None

------- return path[::-1]
```

4.9. Tree Isomorphism.

```
from collections import deque

def bfs(u, adj, n):
--- vis = [False] * n
--- pre = [-1] * n
--- q = deque()

--- q.append(u)
--- vis[u] = True

--- last = u

--- while q:
------- u = q.popleft()
------- last = u

------- for v in adj[u]:
----------- if not vis[v]:
--------------- vis[v] = True
--------------- pre[v] = u
--------------- q.append(v)

--- return last, pre

def tree_centers(r, adj, n):
--- u, _ = bfs(r, adj, n)
--- v, pre = bfs(u, adj, n)

--- path = []
--- while v != -1:
------- path.append(v)
------- v = pre[v]

--- size = len(path)

--- med = [path[size // 2]]
--- if size % 2 == 0:
------- med.append(path[size // 2 - 1])

--- return med

def rootcode(u, adj, p, d, pr):
```

```
    children = []
    for v in adj[u]:
        if v != p:
            children.append(rootcode(v, adj, u, (d + 1) % len(pr), pr))

    children.sort()

    h = len(children) + 1
    for x in children:
        h = h * pr[d] + x

    return h

def treecode(root, adj, n, pr):
    c = tree_centers(root, adj, n)

    if len(c) == 1:
        return (rootcode(c[0], adj, -1, 0, pr) << 1) | 1
    else:
        return (rootcode(c[0], adj, -1, 0, pr) *
                rootcode(c[1], adj, -1, 0, pr)) << 1

def isomorphic(r1, adj1, r2, adj2, n1, n2, pr, rooted=False):
    if rooted:
        return rootcode(r1, adj1, -1, 0, pr) == rootcode(r2, adj2, -1,
↪   0, pr)
    return treecode(r1, adj1, n1, pr) == treecode(r2, adj2, n2, pr) --
```

4.10. Centroid Decomposition.

```
import sys
sys.setrecursionlimit(10**7)

INF = 10**9

class CentroidDecomposition:
    def __init__(self, n):
        self.n = n
        self.adj = [[] for _ in range(n)]

        self.LG = 20
        self.jmp = [[-1] * self.LG for _ in range(n)]
        self.path = [[INF] * self.LG for _ in range(n)]

        self.sz = [0] * n
        self.seph = [0] * n
        self.shortest = [INF] * n
        self.blocked = [False] * n

    def add_edge(self, a, b):
        self.adj[a].append(b)
        self.adj[b].append(a)

    def dfs(self, u, p):
        self.sz[u] = 1
        for v in self.adj[u]:
            if v != p and not self.blocked[v]:
                self.sz[u] += self.dfs(v, u)
        return self.sz[u]

    def find_centroid(self, u, p, n):
        for v in self.adj[u]:
            if v != p and not self.blocked[v] and self.sz[v] > n // 2:
                return self.find_centroid(v, u, n)
        return u

    def makepaths(self, sep, u, p, depth, h):
        self.jmp[u][h] = sep
        self.path[u][h] = depth

        for v in self.adj[u]:
            if v != p and not self.blocked[v]:
                self.makepaths(sep, v, u, depth + 1, h)

    def separate(self, h=0, u=0):
        self.dfs(u, -1)
        c = self.find_centroid(u, -1, self.sz[u])

        self.seph[c] = h
        self.makepaths(c, c, -1, 0, h)

        self.blocked[c] = True

        for v in self.adj[c]:
            if not self.blocked[v]:
                self.separate(h + 1, v)

    def paint(self, u):
        for h in range(self.seph[u] + 1):
            c = self.jmp[u][h]
            self.shortest[c] = min(self.shortest[c], self.path[u][h])

    def closest(self, u):
        ans = INF
        for h in range(self.seph[u] + 1):
            c = self.jmp[u][h]
            ans = min(ans, self.shortest[c] + self.path[u][h])
        return ans
```

5. Math I Algebra

5.0.1. *Generating Function Manager.*

```
MOD = 998244353

def mod_pow(base, exp):
    res = 1
    base %= MOD
    while exp:
        if exp & 1:
            res = res * base % MOD
        base = base * base % MOD
        exp >>= 1
    return res

class GF_Manager:
    DEPTH = 23

    def __init__(self):
        self.prim = [0] * (self.DEPTH + 1)
        self.prim_inv = [0] * (self.DEPTH + 1)
        self.two_inv = [0] * (self.DEPTH + 1)
        self.set_up_primitives()

    def set_up_primitives(self):
        self.prim[self.DEPTH] = 31
        self.prim_inv[self.DEPTH] = mod_pow(self.prim[self.DEPTH], MOD
↪   - 2)
        self.two_inv[self.DEPTH] = mod_pow(1 << self.DEPTH, MOD - 2) -

        for n in range(self.DEPTH - 1, -1, -1):
            self.prim[n] = self.prim[n + 1] * self.prim[n + 1] % MOD -
            self.prim_inv[n] = mod_pow(self.prim[n], MOD - 2)
            self.two_inv[n] = mod_pow(1 << n, MOD - 2)

    def NTT(self, A, n, is_inverse=False):
        if n == 0:
            return A[:]

        half = 1 << (n - 1)
        even = A[0::2]
        odd = A[1::2]

        even = self.NTT(even, n - 1, is_inverse)
        odd = self.NTT(odd, n - 1, is_inverse)

        res = [0] * (1 << n)
        w1 = self.prim_inv[n] if is_inverse else self.prim[n]
        w = 1

        for i in range(half):
            t = w * odd[i] % MOD
            res[i] = (even[i] + t) % MOD
            res[i + half] = (even[i] - t) % MOD
            w = w * w1 % MOD

        return res

    def mult(self, A, B):
        an, bn = len(A), len(B)
        degree = an + bn - 1

        n = 0
        while (1 << n) < degree:
            n += 1

        size = 1 << n
        tA = A + [0] * (size - an)
        tB = B + [0] * (size - bn)

        tA = self.NTT(tA, n)
        tB = self.NTT(tB, n)

        for i in range(size):
            tA[i] = tA[i] * tB[i] % MOD

        tA = self.NTT(tA, n, True)

        inv = self.two_inv[n]
        return [(x * inv) % MOD for x in tA[:degree]]

    def horners(self, F, x):
        ans = 0
        for coef in reversed(F):
            ans = (ans * x + coef) % MOD
        return ans
```

5.0.2. *Fast Fourier Transform.*

```
import math

class Poly:
    def __init__(self, a=0.0, b=0.0):
        self.a = a
        self.b = b

    def __add__(self, other):
        return Poly(self.a + other.a, self.b + other.b)

    def __sub__(self, other):
        return Poly(self.a - other.a, self.b - other.b)

    def __mul__(self, other):
        return Poly(
            self.a * other.a - self.b * other.b,
            self.a * other.b + self.b * other.a
        )

    def __repr__(self):
        return f"({self.a}+{self.b}i)"

def fft_recursive(inp, n, s):
    if n == 1:
        return [inp[0]]
```

```
--- half = n // 2 ----------------------------------------------------------
--------------------------------------------------------------------------------
--- even = fft_recursive(inp, half, s * 2) ---------------------------------
--- odd = fft_recursive(inp[s:], half, s * 2) ------------------------------
--------------------------------------------------------------------------------
--- res = [None] * n -------------------------------------------------------
--- w = Poly(1, 0) ---------------------------------------------------------
--- wn = Poly(math.cos(math.pi / half), math.sin(math.pi / half)) ----
--------------------------------------------------------------------------------
--- for i in range(half): --------------------------------------------------
------- t = w * odd[i] ---------------------------------------------------
------- res[i] = even[i] + t ---------------------------------------------
------- res[i + half] = even[i] - t --------------------------------------
------- w = w * wn -------------------------------------------------------
--------------------------------------------------------------------------------
--- return res -------------------------------------------------------------
--------------------------------------------------------------------------------
def fft(p): ------------------------------------------------------------------
--- return fft_recursive(p, len(p), 1) -------------------------------------
--------------------------------------------------------------------------------
def inverse_fft(p): ----------------------------------------------------------
--- n = len(p) -------------------------------------------------------------
--------------------------------------------------------------------------------
--- p = [Poly(x.a, -x.b) for x in p] ---------------------------------------
--------------------------------------------------------------------------------
--- p = fft(p) -------------------------------------------------------------
--------------------------------------------------------------------------------
--- res = [] ---------------------------------------------------------------
--- for x in p: ------------------------------------------------------------
------- res.append(Poly(x.a / n, -x.b / n)) ------------------------------
--------------------------------------------------------------------------------
--- return res -------------------------------------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
```

5.0.3. *Fft Polynomial Multiplication.*

```
def fft_multiply(a, b): ------------------------------------------------------
--- an, bn = len(a), len(b) ------------------------------------------------
--- degree = an + bn - 1 ---------------------------------------------------
--------------------------------------------------------------------------------
--- n = 1 ------------------------------------------------------------------
--- while n < degree: ------------------------------------------------------
------- n <<= 1 ----------------------------------------------------------
--------------------------------------------------------------------------------
--- A = [Poly(x, 0) for x in a] + [Poly(0, 0)] * (n - an) ------------
--- B = [Poly(x, 0) for x in b] + [Poly(0, 0)] * (n - bn) ------------
--------------------------------------------------------------------------------
--- A = fft(A) -------------------------------------------------------------
--- B = fft(B) -------------------------------------------------------------
--------------------------------------------------------------------------------
--- for i in range(n): -----------------------------------------------------
------- A[i] = A[i] * B[i] -----------------------------------------------
--------------------------------------------------------------------------------
--- A = inverse_fft(A) -----------------------------------------------------
--------------------------------------------------------------------------------
--- result = [int(round(A[i].a)) for i in range(degree)] -------------
--- return result ----------------------------------------------------------
```

5.0.4. *Number Theoretic Transform.*

```
MOD = 998244353 --------------------------------------------------------------
PRIMITIVE_ROOT = 3 -----------------------------------------------------------
--------------------------------------------------------------------------------
def mod_pow(base, exp, mod=MOD): ---------------------------------------------
--- res = 1 ----------------------------------------------------------------
--- base %= mod ------------------------------------------------------------
--- while exp: -------------------------------------------------------------
------- if exp & 1: ------------------------------------------------------
----------- res = res * base % mod -----------------------------------------
------- base = base * base % mod -----------------------------------------
------- exp >>= 1 --------------------------------------------------------
--- return res -------------------------------------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------------------------------------
class Num: -------------------------------------------------------------------
--- def __init__(self, x=0): -----------------------------------------------
------- self.x = x % MOD -------------------------------------------------
--------------------------------------------------------------------------------
--- def __add__(self, other): ----------------------------------------------
------- return Num(self.x + other.x) -------------------------------------
--------------------------------------------------------------------------------
--- def __sub__(self, other): ----------------------------------------------
------- return Num(self.x - other.x) -------------------------------------
--------------------------------------------------------------------------------
--- def __mul__(self, other): ----------------------------------------------
------- return Num(self.x * other.x) -------------------------------------
--------------------------------------------------------------------------------
--- def __truediv__(self, other): ------------------------------------------
------- return self * other.inv() ----------------------------------------
--------------------------------------------------------------------------------
--- def inv(self): ---------------------------------------------------------
------- return Num(mod_pow(self.x, MOD - 2)) -----------------------------
--------------------------------------------------------------------------------
--- def pow(self, p): ------------------------------------------------------
------- return Num(mod_pow(self.x, p)) -----------------------------------
--------------------------------------------------------------------------------
--- def __repr__(self): ----------------------------------------------------
------- return str(self.x) -----------------------------------------------
--------------------------------------------------------------------------------
def ntt(a, invert=False): ----------------------------------------------------
--- n = len(a) -------------------------------------------------------------
--------------------------------------------------------------------------------
--- j = 0 ------------------------------------------------------------------
--- for i in range(1, n): --------------------------------------------------
------- bit = n >> 1 -----------------------------------------------------
------- while j & bit: ---------------------------------------------------
----------- j ^= bit -------------------------------------------------------
----------- bit >>= 1 ------------------------------------------------------
------- j |= bit ----------------------------------------------------------
------- if i < j: --------------------------------------------------------
----------- a[i], a[j] = a[j], a[i] ----------------------------------------
--------------------------------------------------------------------------------
--- length = 2 -------------------------------------------------------------
--- while length <= n: -----------------------------------------------------
------- wlen = mod_pow(PRIMITIVE_ROOT, (MOD - 1) // length) -------
------- if invert: -------------------------------------------------------
----------- wlen = mod_pow(wlen, MOD - 2) ----------------------------------
--------------------------------------------------------------------------------
------- for i in range(0, n, length): ------------------------------------
----------- w = 1 ----------------------------------------------------------
----------- half = length // 2 ---------------------------------------------
----------- for j in range(i, i + half): -----------------------------------
--------------- u = a[j] -------------------------------------------------
--------------- v = Num(a[j + half].x * w % MOD) -------------------------
--------------- a[j] = u + v ---------------------------------------------
--------------- a[j + half] = u - v --------------------------------------
--------------- w = w * wlen % MOD ---------------------------------------
--------------------------------------------------------------------------------
------- length <<= 1 -----------------------------------------------------
--------------------------------------------------------------------------------
--- if invert: -------------------------------------------------------------
------- inv_n = mod_pow(n, MOD - 2) --------------------------------------
------- for i in range(n): -----------------------------------------------
----------- a[i] = Num(a[i].x * inv_n % MOD) -------------------------------
--------------------------------------------------------------------------------
def multiply(A, B): ----------------------------------------------------------
--- n = 1 ------------------------------------------------------------------
--- while n < len(A) + len(B): ---------------------------------------------
------- n <<= 1 ----------------------------------------------------------
--------------------------------------------------------------------------------
--- fa = [Num(x) for x in A] + [Num(0)] * (n - len(A)) ---------------
--- fb = [Num(x) for x in B] + [Num(0)] * (n - len(B)) ---------------
--------------------------------------------------------------------------------
--- ntt(fa) ----------------------------------------------------------------
--- ntt(fb) ----------------------------------------------------------------
--------------------------------------------------------------------------------
--- for i in range(n): -----------------------------------------------------
------- fa[i] = fa[i] * fb[i] --------------------------------------------
--------------------------------------------------------------------------------
--- ntt(fa, True) ----------------------------------------------------------
--------------------------------------------------------------------------------
--- return [fa[i].x for i in range(len(A) + len(B) - 1)] -------------
--------------------------------------------------------------------------------
def poly_inv(x, l): ----------------------------------------------------------
--- if l == 1: -------------------------------------------------------------
------- return [mod_pow(x[0], MOD - 2)] ----------------------------------
--------------------------------------------------------------------------------
--- y = poly_inv(x, (l + 1) // 2) ------------------------------------------
--------------------------------------------------------------------------------
--- n = 1 ------------------------------------------------------------------
--- while n < 2 * l: -------------------------------------------------------
------- n <<= 1 ----------------------------------------------------------
--------------------------------------------------------------------------------
--- A = x[:l] + [0] * (n - l) ----------------------------------------------
--- B = y[:] + [0] * (n - len(y)) ------------------------------------------
--------------------------------------------------------------------------------
--- A = [Num(v) for v in A] ------------------------------------------------
--- B = [Num(v) for v in B] ------------------------------------------------
--------------------------------------------------------------------------------
--- ntt(A) -----------------------------------------------------------------
--- ntt(B) -----------------------------------------------------------------
--------------------------------------------------------------------------------
--- for i in range(n): -----------------------------------------------------
------- B[i] = B[i] * (Num(2) - A[i] * B[i]) -----------------------------
--------------------------------------------------------------------------------
--- ntt(B, True) -----------------------------------------------------------
--------------------------------------------------------------------------------
--- return [B[i].x % MOD for i in range(l)] ---------------------------
--------------------------------------------------------------------------------
def poly_sqrt(x, l): ---------------------------------------------------------
--- if l == 1: -------------------------------------------------------------
------- assert x[0] == 1 -------------------------------------------------
------- return [1] -------------------------------------------------------
--------------------------------------------------------------------------------
--- y = poly_sqrt(x, l // 2) -----------------------------------------------
--- inv_y = poly_inv(y, l // 2) --------------------------------------------
--------------------------------------------------------------------------------
--- n = 1 ------------------------------------------------------------------
--- while n < 2 * l: -------------------------------------------------------
------- n <<= 1 ----------------------------------------------------------
--------------------------------------------------------------------------------
--- T1 = x[:l] + [0] * (n - l) ---------------------------------------------
--- T2 = inv_y[:] + [0] * (n - len(inv_y)) ---------------------------------
--------------------------------------------------------------------------------
--- T1 = [Num(v) for v in T1] ----------------------------------------------
--- T2 = [Num(v) for v in T2] ----------------------------------------------
--------------------------------------------------------------------------------
--- ntt(T1) ----------------------------------------------------------------
--- ntt(T2) ----------------------------------------------------------------
--------------------------------------------------------------------------------
--- for i in range(n): -----------------------------------------------------
------- T2[i] = T1[i] * T2[i] --------------------------------------------
--------------------------------------------------------------------------------
--- ntt(T2, True) ----------------------------------------------------------
--------------------------------------------------------------------------------
--- inv2 = (MOD + 1) // 2 --------------------------------------------------
--- res = [] ---------------------------------------------------------------
--- for i in range(l): -----------------------------------------------------
------- val = (y[i] + T2[i].x) * inv2 % MOD ------------------------------
------- res.append(val) --------------------------------------------------
--------------------------------------------------------------------------------
--- return res -------------------------------------------------------------
```

5.0.5. *Polynomial Long Division.*

```
MOD = 998244353
G = 3

def mod_pow(a, e):
    r = 1
    while e:
        if e & 1:
            r = r * a % MOD
        a = a * a % MOD
        e >>= 1
    return r

def bit_reverse(a):
    n = len(a)
    j = 0
    for i in range(1, n):
        bit = n >> 1
        while j & bit:
            j ^= bit
            bit >>= 1
        j ^= bit
        if i < j:
            a[i], a[j] = a[j], a[i]

def ntt(a, invert=False):
    n = len(a)
    bit_reverse(a)

    length = 2
    while length <= n:
        wlen = mod_pow(G, (MOD - 1) // length)
        if invert:
            wlen = mod_pow(wlen, MOD - 2)

        for i in range(0, n, length):
            w = 1
            half = length // 2
            for j in range(i, i + half):
                u = a[j]
                v = a[j + half] * w % MOD
                a[j] = (u + v) % MOD
                a[j + half] = (u - v) % MOD
                w = w * wlen % MOD

        length <<= 1

    if invert:
        inv_n = mod_pow(n, MOD - 2)
        for i in range(n):
            a[i] = a[i] * inv_n % MOD

def multiply(a, b):
    n = 1
    while n < len(a) + len(b):
        n <<= 1

    fa = a[:] + [0] * (n - len(a))
    fb = b[:] + [0] * (n - len(b))

    ntt(fa)
    ntt(fb)

    for i in range(n):
        fa[i] = fa[i] * fb[i] % MOD

    ntt(fa, True)

    return fa[:len(a) + len(b) - 1]

def poly_inv(f, n):
    if n == 1:
        return [mod_pow(f[0], MOD - 2)]

    m = (n + 1) // 2
    g = poly_inv(f, m)

    size = 1
    while size < 2 * n:
        size <<= 1

    a = f[:n] + [0] * (size - len(f[:n]))
    g = g + [0] * (size - len(g))

    ntt(a)
    ntt(g)

    for i in range(size):
        g[i] = g[i] * (2 - a[i] * g[i] % MOD) % MOD

    ntt(g, True)

    return g[:n]

def poly_divide(A, B):
    n, m = len(A), len(B)

    if n < m:
        return [], A[:]

    k = n - m + 1

    A_rev = A[::-1]
    B_rev = B[::-1]

    B_inv = poly_inv(B_rev, k)

    Q_rev = multiply(A_rev, B_inv)[:k]
    Q = Q_rev[::-1]

    BQ = multiply(B, Q)

    R = [(A[i] - (BQ[i] if i < len(BQ) else 0)) % MOD for i in
    ↪  range(n)]

    while Q and Q[-1] == 0:
        Q.pop()
    while R and R[-1] == 0:
        R.pop()

    return Q, R
```

5.0.6. *Matrix Multiplication.*

```
MOD = 998244353

def multiply(A, B):
    p = len(A)
    q = len(A[0])
    r = len(B[0])

    if len(B) != q:
        raise ValueError("Dimension mismatch")

    AB = [[0] * r for _ in range(p)]

    for i in range(p):
        for j in range(q):
            for k in range(r):
                AB[i][k] = (AB[i][k] + A[i][j] * B[j][k]) % MOD

    return AB
```

5.0.7. *Matrix Power.*

```
MOD = 998244353

def multiply(A, B):
    p, q, r = len(A), len(A[0]), len(B[0])

    res = [[0] * r for _ in range(p)]

    for i in range(p):
        for j in range(q):
            for k in range(r):
                res[i][k] = (res[i][k] + A[i][j] * B[j][k]) % MOD

    return res

def identity(n):
    I = [[0] * n for _ in range(n)]
    for i in range(n):
        I[i][i] = 1
    return I

def power(B, e):
    n = len(B)

    result = identity(n)
    base = [row[:] for row in B]

    while e > 0:
        if e & 1:
            result = multiply(result, base)
        base = multiply(base, base)
        e >>= 1

    return result
```

5.0.8. *Gauss Jordan Matrix Determinant.*

```
EPS = 1e-9

def gauss_jordan(A):
    n = len(A)
    m = len(A[0])

    singular = False

    i = 0
    for p in range(m):
        if i >= n:
            break

        pivot = i
        for k in range(i + 1, n):
            if abs(A[k][p]) > abs(A[pivot][p]):
                pivot = k

        if abs(A[pivot][p]) < EPS:
            singular = True
            continue

        A[i], A[pivot] = A[pivot], A[i]

        div = A[i][p]
        for j in range(p, m):
            A[i][j] /= div

        for k in range(n):
            if k == i:
                continue
            factor = A[k][p]
            for j in range(p, m):
                A[k][j] -= factor * A[i][j]

        i += 1
```

```
--- return not singular
def apply(A, x):
--- return [sum(A[i][j] * x[j] for j in range(len(x))) for i in
↪  range(len(A))]
```

6. Math II Combinatorics

6.0.1. *Lucas Theorem.*

```
_fact_cache = []
_last_p = -1

def lucas(n, k, p):
--- if k < 0 or k > n:
------- return 0
--- if k == 0 or n == k:
------- return 1

--- global _fact_cache, _last_p
--- if _last_p != p:
------- _fact_cache = [1] * p
------- for i in range(2, p):
----------- _fact_cache[i] = (_fact_cache[i - 1] * i) % p
------- _last_p = p

--- res_num = 1
--- res_den = 1

--- while n > 0 or k > 0:
------- ni, ki = n % p, k % p
------- if ki > ni:
----------- return 0
------- res_num = (res_num * _fact_cache[ni]) % p
------- res_den = (res_den * _fact_cache[ki] * _fact_cache[ni - ki]) %
↪  p
------- n //= p
------- k //= p
--- return (res_num * pow(res_den, p - 2, p)) % p
# Input Format: print(lucas(n, k, p))
```

6.0.2. *Catalan Numbers.*

```
def catalan(n):
--- if (n == 0 or n == 1):
------- return 1

--- dp = [0] * (n + 1)
--- dp[0] = dp[1] = 1

--- for i in range(2, n + 1):
------- for j in range(i):
----------- dp[i] += dp[j] * dp[i-j-1]

--- return dp[n]

# for i in range(10):
#     print(catalan(i))
```

6.0.3. *Kth Permutation.*

```
#fact = [1, 1, 2, 6, 24, 120, 720, 5040, 40320, 362880]

def findKthPermutation(n, k):
--- nums = [str(i) for i in range(1, n + 1)]

--- fact = [1] * n
--- for i in range(1, n):
------- fact[i] = fact[i - 1] * i

--- k -= 1
--- res = []

--- for i in range(n - 1, -1, -1):
------- idx = k // fact[i]
------- k %= fact[i]

------- res.append(nums.pop(idx))

--- return ''.join(res)
```

6.0.4. *Factoradics.*

```
def factoradic(arr):
--- n = len(arr)
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in arr:
------- bit_val = val + 1
------- s = ft.query(bit_val - 1)
------- ft.add(bit_val, -1)
------- res.append(s)
--- return res

def permute(fact, n):
--- ft = FenwickTree(n)
--- for i in range(1, n + 1):
------- ft.add(i, 1)

--- res = []
--- for val in fact:
------- idx = ft.find_kth(val)
------- ft.add(idx, -1)
------- res.append(idx - 1)
--- return res
```

7. Math III Number Theory

7.0.1. *Gcd Subset Counter.*

```
def gcd_subset_count(arr):
--- n = len(arr)
--- mx = max(arr)

--- f = [0] * (mx + 1)
--- for x in arr:
------- f[x] += 1

--- gcnt = [0] * (mx + 1)

--- for i in range(mx, 0, -1):
------- add = 0

------- for j in range(i, mx + 1, i):
----------- add += f[j]

------- total = (1 << add) - 1  # 2^add - 1

------- sub = 0
------- for j in range(2 * i, mx + 1, i):
----------- sub += gcnt[j]

------- gcnt[i] = total - sub

--- return gcnt
```

7.0.2. *Linear Diophantine.*

```
def gcd(a, b):
--- if a == 0:
------- return b
--- return gcd(b % a, a)

def isPossible(a, b, c):
--- g = gcd(a, b)

--- return c % g == 0
```

7.0.3. *Chinese Remainder Theorem.*

```
def gcd_extended(a, b):
--- if a == 0:
------- return b, 0, 1
--- gcd, x1, y1 = gcd_extended(b % a, a)
--- x = y1 - (b // a) * x1
--- y = x1
--- return gcd, x, y

def find_min_x(num, rem):
--- prod = 1
--- for n in num:
------- prod *= n

--- result = 0
--- for i in range(len(num)):
------- prod_i = prod // num[i]
------- _, inv_i, _ = gcd_extended(prod_i, num[i])
------- result += rem[i] * prod_i * inv_i

--- return result % prod
```

7.0.4. *Euler Totient.*

```
def totient(n):
--- if n <= 1:
------- return 1

--- tot = n
--- i = 2

--- while i * i <= n:
------- if n % i == 0:
----------- tot -= tot // i
----------- while n % i == 0:
--------------- n //= i
------- i += 1

--- if n > 1:
------- tot -= tot // n

--- return tot
```

7.0.5. *Extended Euclidean Algorithm.*

```
def mod(x, m):
--- if m == 0:
------- return 0
--- if m < 0:
------- m = -m
--- return (x % m + m) % m

def extended_euclid(a, b):
--- if b == 0:
------- return a, 1, 0

--- g, x1, y1 = extended_euclid(b, a % b)

--- x = y1
--- y = x1 - (a // b) * y1

--- return g, x, y
```

7.0.6. *Modular Exponentiation.*

```
def mod_pow(b, e, m):
    res = 1 % m

    while e > 0:
        if e & 1:
            res = (res * b) % m
        b = (b * b) % m
        e >>= 1

    return res
```

7.0.7. *Mod Inverse.*

```
def extended_euclid(a, b):
    if b == 0:
        return a, 1, 0

    g, x1, y1 = extended_euclid(b, a % b)

    x = y1
    y = x1 - (a // b) * y1

    return g, x, y


def modinv(a, m):
    g, x, y = extended_euclid(a, m)

    if g != 1:
        return 0  # no inverse

    return x % m
```

7.0.8. *Modulo Solver.*

```
def extended_euclid(a, b):
    if b == 0:
        return a, 1, 0

    g, x1, y1 = extended_euclid(b, a % b)

    x = y1
    y = x1 - (a // b) * y1

    return g, x, y


def modsolver(a, b, m):
    g, x, y = extended_euclid(a, m)

    if b % g != 0:
        return (-1, -1)

    m_reduced = m // g

    x0 = (x * (b // g)) % m_reduced

    return (x0, m_reduced)
```

7.0.9. *Primitive Root.*

```
def primitive_root(m):
    if m == 2:
        return 1

    phi = m - 1

    factors = set()
    x = phi
    i = 2

    while i * i <= x:
        if x % i == 0:
            factors.add(i)
            while x % i == 0:
                x //= i
        i += 1

    if x > 1:
        factors.add(x)

    for g in range(2, m):
        ok = True
        for f in factors:
            if mod_pow(g, phi // f, m) == 1:
                ok = False
                break
        if ok:
            return g

    return -1
```

7.0.10. *Josephus.*

```
def josephus(n, k):
    res = 0
    for i in range(2, n + 1):
        res = (res + k) % i
    return res
```

8. Math IV Numerical Methods

8.0.1. *Fast Square Testing.*

```
import math

class FastSquareTester:
    def __init__(self):
        self.M = 0
        self.init_is_square()

    def init_is_square(self):
        for i in range(64):
            self.M |= (1 << (i * i % 64))

    def is_square(self, x):
        if x < 0: return False
        if x == 0: return True

        if not (self.M & (1 << (x & 63))):
            return False

        c = (x & -x).bit_length() - 1
        if c & 1:
            return False

        temp_x = x >> c
        if (temp_x & 7) != 1:
            return False

        r = int(math.isqrt(temp_x))
        return r * r == temp_x

tester = FastSquareTester()
```

8.0.2. *Simpson Integration.*

```
def f(x):
    return x ** 2
def simpson_integration(a, b, n=1000000):
    if n % 2 != 0:
        n += 1

    h = (b - a) / n
    s = f(a) + f(b)

    for i in range(1, n):
        x = a + h * i
        if i % 2 == 1:
            s += 4 * f(x)
        else:
            s += 2 * f(x)

    return s * (h / 3)
```

9. Strings

9.0.1. *Longest Common Prefix.*

```
def build_lcp(s, sa):
    n = len(s)
    rank = [0] * n
    lcp = [0] * n

    for i in range(n):
        rank[sa[i]] = i

    k = 0
    for i in range(n):
        if rank[i] == 0:
            k = 0
            continue

        j = sa[rank[i] - 1]

        while i + k < n and j + k < n and s[i + k] == s[j + k]:
            k += 1

        lcp[rank[i]] = k

        if k > 0:
            k -= 1

    return lcp
```

9.0.2. *Hasher.*

```
class Hasher:
    def __init__(self, s, bases=[31, 37], mod=10 ** 9 + 7):
        self.bases = bases
        self.mod = mod
        self.n = len(s)
        self.num_hashes = len(bases)
        self.p_pow = [[1] * (self.n + 1) for _ in range(self.num_hashes)]
        self.h_ans = [[0] * (self.n + 1) for _ in range(self.num_hashes)]

        for i, b in enumerate(self.bases):
            for j in range(self.n):
                self.p_pow[i][j + 1] = (self.p_pow[i][j] * b) % self.mod
                self.h_ans[i][j + 1] = (self.h_ans[i][j] + s[j] * self.p_pow[i][j]) % self.mod

    def get_hash(self, l, r):
        return tuple((self.h_ans[i][r + 1] - self.h_ans[i][l]) % self.mod
                     for i in range(self.num_hashes))

    def are_equal(self, l1, r1, l2, r2):
        if (r1 - l1) != (r2 - l2): return False

        h1 = self.get_hash(l1, r1)
        h2 = self.get_hash(l2, r2)

        return all((h1[i] * self.p_pow[i][l2]) % self.mod ==
```

```
------------------ (h2[i] * self.p_pow[i][l1]) % self.mod ------------
------------------ for i in range(self.num_hashes)) ------------------
```

9.0.3. *Z Algorithm.*

9.0.4. *Aho Corasick Trie.*

```
from collections import deque ----------------------------------------
import random --------------------------------------------------------
import string --------------------------------------------------------
 ---------------------------------------------------------------------
class Node: ----------------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.next = {} ---------------------------------------------
------- self.fail = None -------------------------------------------
------- self.count = 0 ---------------------------------------------
 ---------------------------------------------------------------------
--- def add(self, s): --------------------------------------------
------- node = self ------------------------------------------------
------- for c in s: ------------------------------------------------
----------- if c not in node.next: -------------------------------
--------------- node.next[c] = Node() ------------------------------
----------- node = node.next[c] ----------------------------------
------- node.count += 1 --------------------------------------------
 ---------------------------------------------------------------------
--- def prepare(self): -------------------------------------------
------- root = self ------------------------------------------------
------- q = deque() ------------------------------------------------
 ---------------------------------------------------------------------
------- root.fail = root -------------------------------------------
 ---------------------------------------------------------------------
------- for child in root.next.values(): ---------------------------
----------- child.fail = root ------------------------------------
----------- q.append(child) --------------------------------------
 ---------------------------------------------------------------------
------- while q: ---------------------------------------------------
----------- head = q.popleft() -----------------------------------
 ---------------------------------------------------------------------
----------- for c, nxt in head.next.items(): ---------------------
--------------- f = head.fail ---------------------------------------
 ---------------------------------------------------------------------
--------------- while f != root and c not in f.next: ----------------
------------------- f = f.fail ------------------------------------
 ---------------------------------------------------------------------
--------------- nxt.fail = f.next[c] if c in f.next else root --------
 ---------------------------------------------------------------------
--------------- nxt.count += nxt.fail.count --------------------------
 ---------------------------------------------------------------------
--------------- q.append(nxt) ----------------------------------------
 ---------------------------------------------------------------------
--- def search(self, s): ----------------------------------------
------- root = self ------------------------------------------------
------- p = self ---------------------------------------------------
------- ans = 0 ----------------------------------------------------
 ---------------------------------------------------------------------
------- for c in s: ------------------------------------------------
----------- while p != root and c not in p.next: -----------------
--------------- p = p.fail ------------------------------------------
 ---------------------------------------------------------------------
----------- if c in p.next: --------------------------------------
--------------- p = p.next[c] ---------------------------------------
--------------- ans += p.count --------------------------------------
 ---------------------------------------------------------------------
------- return ans -------------------------------------------------
```

9.0.5. *Knuth Morris Pratt.*

```
def build_kmp(f): ----------------------------------------------------
--- n = len(f) ---------------------------------------------------
--- par = [0] * (n + 1) ------------------------------------------
 ---------------------------------------------------------------------
--- par[0] = -1 --------------------------------------------------
--- if n > 0: ----------------------------------------------------
------- par[1] = 0 -------------------------------------------------
 ---------------------------------------------------------------------
--- i, j = 2, 0 --------------------------------------------------
 ---------------------------------------------------------------------
--- while i <= n: ------------------------------------------------
------- if f[i - 1] == f[j]: ---------------------------------------
----------- j += 1 -----------------------------------------------
----------- par[i] = j -------------------------------------------
----------- i += 1 -----------------------------------------------
------- elif j > 0: ------------------------------------------------
----------- j = par[j] -------------------------------------------
------- else: ------------------------------------------------------
----------- par[i] = 0 -------------------------------------------
----------- i += 1 -----------------------------------------------
 ---------------------------------------------------------------------
--- return par ---------------------------------------------------
 ---------------------------------------------------------------------
 ---------------------------------------------------------------------
def kmp(s, f): -------------------------------------------------------
--- par = build_kmp(f) -------------------------------------------
 ---------------------------------------------------------------------
--- i = 0 --------------------------------------------------------
--- j = 0 --------------------------------------------------------
--- ans = [] -----------------------------------------------------
 ---------------------------------------------------------------------
--- n, m = len(s), len(f) ----------------------------------------
 ---------------------------------------------------------------------
--- while i + j < n: ---------------------------------------------
------- if s[i + j] == f[j]: ---------------------------------------
----------- j += 1 -----------------------------------------------
----------- if j == m: -------------------------------------------
--------------- ans.append(i) ---------------------------------------
--------------- i += j - par[j] -------------------------------------
--------------- if j > 0: -------------------------------------------
------------------ j = par[j] -------------------------------------
------- else: ------------------------------------------------------
----------- i += j - par[j] --------------------------------------
----------- if j > 0: --------------------------------------------
--------------- j = par[j] ------------------------------------------
 ---------------------------------------------------------------------
--- return ans ---------------------------------------------------
```

9.0.6. *Trie.*

```
class Node: ----------------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.children = {} -----------------------------------------
------- self.prefixes = 0 ------------------------------------------
------- self.words = 0 ---------------------------------------------
 ---------------------------------------------------------------------
class Trie: ----------------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.root = Node() -----------------------------------------
 ---------------------------------------------------------------------
--- def insert(self, seq): ---------------------------------------
------- cur = self.root --------------------------------------------
 ---------------------------------------------------------------------
------- for i in range(len(seq) + 1): ------------------------------
----------- cur.prefixes += 1 ------------------------------------
 ---------------------------------------------------------------------
----------- if i == len(seq): ------------------------------------
--------------- cur.words += 1 --------------------------------------
--------------- break -----------------------------------------------
 ---------------------------------------------------------------------
----------- head = seq[i] ----------------------------------------
 ---------------------------------------------------------------------
----------- if head not in cur.children: -------------------------
--------------- cur.children[head] = Node() -------------------------
 ---------------------------------------------------------------------
----------- cur = cur.children[head] -----------------------------
 ---------------------------------------------------------------------
--- def countMatches(self, seq): ---------------------------------
------- cur = self.root --------------------------------------------
 ---------------------------------------------------------------------
------- for c in seq: ----------------------------------------------
----------- if c not in cur.children: ----------------------------
--------------- return 0 -------------------------------------------
----------- cur = cur.children[c] --------------------------------
 ---------------------------------------------------------------------
------- return cur.words -------------------------------------------
 ---------------------------------------------------------------------
--- def countPrefixes(self, seq): ------------------------------------
------- cur = self.root --------------------------------------------
 ---------------------------------------------------------------------
------- for c in seq: ----------------------------------------------
----------- if c not in cur.children: ----------------------------
--------------- return 0 -------------------------------------------
----------- cur = cur.children[c] --------------------------------
 ---------------------------------------------------------------------
------- return cur.prefixes ----------------------------------------
```

9.0.7. *Suffix Array.*

```
class SuffixArray: ---------------------------------------------------
--- def __init__(self): ------------------------------------------
------- self.n = 0 -------------------------------------------------
------- self.suffix = [] -------------------------------------------
------- self.equiv = [] --------------------------------------------
------- self.equiv_pair = [] ---------------------------------------
------- self.s = "" ------------------------------------------------
 ---------------------------------------------------------------------
--- def build(self, s): ------------------------------------------
------- if not s.endswith('$'): ------------------------------------
----------- s += '$' ---------------------------------------------
 ---------------------------------------------------------------------
------- self.s = s -------------------------------------------------
------- self.n = len(s) --------------------------------------------
 ---------------------------------------------------------------------
------- self.suffix = list(range(self.n)) --------------------------
------- self.equiv = [0] * self.n ----------------------------------
------- self.equiv_pair = [(0, 0)] * self.n ------------------------
 ---------------------------------------------------------------------
------- self.suffix.sort(key=lambda i: s[i]) -----------------------
 ---------------------------------------------------------------------
------- cls = 0 ----------------------------------------------------
------- for i in range(self.n): ------------------------------------
----------- if i == 0 or s[self.suffix[i]] != s[self.suffix[i - 1]]: -
--------------- cls += 1 --------------------------------------------
----------- self.equiv[self.suffix[i]] = cls ---------------------
 ---------------------------------------------------------------------
------- k = 1 ------------------------------------------------------
------- while k < self.n: ------------------------------------------
----------- for i in range(self.n): ------------------------------
--------------- self.equiv_pair[i] = ( ------------------------------
------------------ self.equiv[i], -------------------------------
------------------ self.equiv[(i + k) % self.n] -----------------
--------------- ) ---------------------------------------------------
 ---------------------------------------------------------------------
----------- self.suffix.sort(key=lambda i: self.equiv_pair[i]) -------
 ---------------------------------------------------------------------
----------- cls = 0 ----------------------------------------------
----------- new_equiv = [0] * self.n -----------------------------
 ---------------------------------------------------------------------
----------- for i in range(self.n): ------------------------------
--------------- if i == 0 or self.equiv_pair[self.suffix[i]] !=
↪  self.equiv_pair[self.suffix[i - 1]]:
------------------ cls += 1 -------------------------------------
--------------- new_equiv[self.suffix[i]] = cls ---------------------
 ---------------------------------------------------------------------
----------- self.equiv = new_equiv -------------------------------
----------- k <<= 1 ----------------------------------------------
 ---------------------------------------------------------------------
--- def _cmp(self, start, pattern): ------------------------------
```

```
------- s = self.s
------- n = len(s)

------- i = start
------- j = 0

------- while i < n and j < len(pattern):
----------- if s[i] != pattern[j]:
--------------- return ord(s[i]) - ord(pattern[j])
----------- i += 1
----------- j += 1

------- if j == len(pattern):
----------- return 0
------- return -1

--- def lower_bound(self, pattern):
------- l, r = 0, self.n - 1
------- res = self.n

------- while l <= r:
----------- m = (l + r) // 2
----------- if self._cmp(self.suffix[m], pattern) >= 0:
--------------- res = m
--------------- r = m - 1
----------- else:
--------------- l = m + 1

------- return res

--- def upper_bound(self, pattern):
------- l, r = 0, self.n - 1
------- res = self.n

------- while l <= r:
----------- m = (l + r) // 2
----------- if self._cmp(self.suffix[m], pattern) > 0:
--------------- res = m
--------------- r = m - 1
----------- else:
--------------- l = m + 1

------- return res

--- def count_occurrences(self, pattern):
------- L = self.lower_bound(pattern)
------- R = self.upper_bound(pattern)
------- return max(0, R - L)
```

9.0.8. *Booth Min String Rotation.*

```
def booth(s: str) -> int:
--- s = s + s
--- n = len(s)

--- f = [-1] * n
--- k = 0

--- for j in range(1, n):
------- i = f[j - k - 1]

------- while i != -1 and s[j] != s[k + i + 1]:
----------- if s[j] < s[k + i + 1]:
--------------- k = j - i - 1
----------- i = f[i]

------- if i == -1 and s[j] != s[k]:
----------- if s[j] < s[k]:
--------------- k = j
----------- f[j - k] = -1
------- else:
----------- f[j - k] = i + 1

--- return k
```

10. Other Algorithms

10.0.1. *Longest Increasing Subsequence.*

```
def lis(arr):
--- if not arr:
------- return []

--- n = len(arr)
--- seq = []
--- back = [0] * n

--- for i in range(n):
------- lo, hi = 0, len(seq)
------- while lo < hi:
----------- mid = (lo + hi) // 2
----------- if arr[seq[mid]] < arr[i]:
--------------- lo = mid + 1
----------- else:
--------------- hi = mid

------- res = lo
------- if res < len(seq):
----------- seq[res] = i
------- else:
----------- seq.append(i)
------- back[i] = -1 if res == 0 else seq[res - 1]

--- ans = []
--- at = seq[-1]
--- while at != -1:
------- ans.append(at)
------- at = back[at]

--- return ans[::-1]
```

10.0.2. *Cycle Find.*

```
def find_cycle(x0, f):
--- t = f(x0)
--- h = f(f(x0))

--- while t != h:
------- t = f(t)
------- h = f(f(h))

--- mu = 0
--- h = x0
--- while t != h:
------- t = f(t)
------- h = f(h)
------- mu += 1

--- lam = 1
--- h = f(t)
--- while t != h:
------- h = f(h)
------- lam += 1

--- return mu, lam
```

10.0.3. *2Sat.*

```
class TwoSAT:
--- def __init__(self, n):
------- self.n = n
------- self.N = 2 * n
------- self.adj = [[] for _ in range(self.N)]

--- def var(self, x):
------- return x << 1

--- def neg(self, x):
------- return x ^ 1

--- def add_imp(self, x, y):
------- self.adj[x].append(y)

--- def add_or(self, x, y):
------- self.add_imp(self.neg(x), y)
------- self.add_imp(self.neg(y), x)

--- def sat(self):
------- n = self.N
------- adj = self.adj

------- index = 0
------- stack = []
------- onstack = [False] * n
------- ids = [-1] * n
------- low = [0] * n
------- comp = [-1] * n
------- self.time = 0
------- self.comp_id = 0

------- def dfs(u):
----------- nonlocal index

----------- ids[u] = low[u] = index
----------- index += 1
----------- stack.append(u)
----------- onstack[u] = True

----------- for v in adj[u]:
--------------- if ids[v] == -1:
------------------- dfs(v)
------------------- low[u] = min(low[u], low[v])
--------------- elif onstack[v]:
------------------- low[u] = min(low[u], ids[v])

----------- if low[u] == ids[u]:
--------------- while True:
------------------- v = stack.pop()
------------------- onstack[v] = False
------------------- comp[v] = self.comp_id
------------------- if v == u:
----------------------- break
--------------- self.comp_id += 1

------- for i in range(n):
----------- if ids[i] == -1:
--------------- dfs(i)

------- for i in range(0, n, 2):
----------- if comp[i] == comp[i ^ 1]:
--------------- return False

------- return True
```

10.0.4. *Dpll.*

```
class SAT:
--- def __init__(self):
------- self.clauses = []
------- self.n = 0

--- def var(self):
------- self.n += 1
------- return self.n

--- def clause(self, c):
------- self.clauses.append(c)
```

```python
    def clause_sat(self, clause, assign):
        unknown = False

        for x in clause:
            v = abs(x)
            val = assign[v]

            if x > 0 and val == 1:
                return True
            if x < 0 and val == 0:
                return True

            if val == -1:
                unknown = True

        return False if not unknown else None

    def propagate(self, assign):
        changed = True

        while changed:
            changed = False

            for clause in self.clauses:
                sat = False
                unassigned = []

                for x in clause:
                    v = abs(x)

                    if x > 0 and assign[v] == 1:
                        sat = True
                        break
                    if x < 0 and assign[v] == 0:
                        sat = True
                        break

                    if assign[v] == -1:
                        unassigned.append(x)

                if sat:
                    continue

                if len(unassigned) == 0:
                    return False  # contradiction

                if len(unassigned) == 1:
                    x = unassigned[0]
                    v = abs(x)
                    val = 1 if x > 0 else 0

                    if assign[v] != -1 and assign[v] != val:
                        return False

                    if assign[v] == -1:
                        assign[v] = val
                        changed = True

        return True

    def solve_dfs(self, assign, idx=1):
        if idx > self.n:
            return True

        if assign[idx] != -1:
            return self.solve_dfs(assign, idx + 1)

        for val in [0, 1]:
            new_assign = assign[:]
            new_assign[idx] = val

            if self.propagate(new_assign):
                if self.solve_dfs(new_assign, idx + 1):
                    return True

        return False

    def solve(self):
        assign = [-1] * (self.n + 1)

        if not self.propagate(assign):
            return False

        return self.solve_dfs(assign)

    def get_value(self, x):
        return self.assign[x] == 1
```

10.0.5. *Stable Marriage.*

```python
from collections import deque

def stable_marriage(n, m_pref, w_pref):
    w_rank = [[0] * n for _ in range(n)]

    for w in range(n):
        for i, man in enumerate(w_pref[w]):
            w_rank[w][man] = i

    q = deque(range(n))

    at = [0] * n

    eng = [-1] * n

    res = [-1] * n

    while q:
        m = q.popleft()

        while at[m] < n:
            w = m_pref[m][at[m]]
            at[m] += 1

            if eng[w] == -1:
                eng[w] = m
                res[m] = w
                break

            if w_rank[w][m] < w_rank[w][eng[w]]:
                q.append(eng[w])
                eng[w] = m
                res[m] = w
                break

    return res
```

10.0.6. *Dates.*

```python
def c_div(a, b):
    return int(a / b)

def date_to_int(y, m, d):
    k = c_div(m - 14, 12)

    return (
        1461 * (y + 4800 + k) // 4
        + 367 * (m - 2 - k * 12) // 12
        - 3 * ((y + 4900 + k) // 100) // 4
        + d - 32075
    )

def int_to_day(jd):
    return jd % 7

def int_to_date(jd):
    x = jd + 68569
    n = 4 * x // 146097
    x -= (146097 * n + 3) // 4

    i = (4000 * (x + 1)) // 1461001
    x -= 1461 * i // 4 - 31

    j = 80 * x // 2447
    d = x - 2447 * j // 80

    x = j // 11
    m = j + 2 - 12 * x
    y = 100 * (n - 49) + i + x

    return y, m, d
```

10.0.7. *Simulated Annealing.*

```python
import random
import math
import time


def simulated_annealing(n, seconds):
    rng = random.Random()

    sol = list(range(1, n + 1))
    rng.shuffle(sol)

    def score(arr):
        return sum(abs(arr[i] - arr[i - 1]) for i in range(1, n))

    cur = score(sol)

    T0 = 100.0
    T1 = 0.001
    start = time.time()
    iters = 0

    while True:
        if iters % 16 == 0:
            progress = (time.time() - start) / seconds
            if progress >= 1.0:
                break
            temp = T0 * (T1 / T0) ** progress

        a = rng.randint(0, n - 2)

        x, y = sol[a], sol[a + 1]

        delta = 0

        if a > 0:
            delta += abs(sol[a - 1] - y) - abs(sol[a - 1] - x)

        if a + 2 < n:
            delta += abs(x - sol[a + 2]) - abs(y - sol[a + 2])

        accept = delta <= 0 or rng.random() < math.exp(-delta / temp)

        if accept:
            sol[a], sol[a + 1] = sol[a + 1], sol[a]
            cur += delta

        iters += 1

    return cur, sol
```

10.0.8. *Fast Input Reading.*

```
import sys ------------------------------------------------------------
 ----------------------------------------------------------------------
data = sys.stdin.buffer.read().split() -------------------------------
it = iter(data) -------------------------------------------------------
 ----------------------------------------------------------------------
def next_int(): -------------------------------------------------------
--- return int(next(it)) ------------------------------------------------
```

10.0.9. *Bit Hacks.*

```
def snoob(x): ---------------------------------------------------------
--- y = x & -x ------------------------------------------------------
--- z = x + y -------------------------------------------------------
--- return z | (((x ^ z) // y) >> 2) --------------------------------
```